Numericals

Example 6.2.1 *For Hg of mass number 202, the T_c value is 4.2 K. Find the T_c value for the isotope of mercury of mass number 199.5 Take $\alpha = 0.5$.*

GTU : Summer-22, Marks 3

Solution :

Given :

$$T_{c1} = 4.2 \text{ K}, \quad M_1 = 202$$

$$T_{c2} = ? \qquad M_2 = 199.5$$

$$\alpha = 0.5$$

Formula,

$$T_c \propto \frac{1}{\sqrt{M}}$$

$$T_c M^{\frac{1}{2}} = \text{Constant}$$

$$\frac{T_{c1}}{T_{c1}} = \frac{\sqrt{M_2}}{\sqrt{M_1}}$$

$$\therefore \quad T_{c2} = \frac{T_{c1} \times \sqrt{M_1}}{\sqrt{M_2}} = \frac{4.2 \times \sqrt{202}}{199.5}$$

$$\therefore \quad T_{c2} = \mathbf{4.22\ K}$$

Example 6.2.3 *Critical magnetic field of lead is 6.5×10^3 A/m at 0 K. Calculate temperature at which the critical magnetic field of lead drops to 3×10^3 A/m. The critical temperature of lead is 7.2 K. Calculate critical current density at that temperature if radius of the wire is 0.5 mm.* **GTU : Summer-22, Marks 4**

Solution :

$$H_c(0) = 6.5 \times 10^3 \text{ A/m}$$

$$T = ?,$$

$$H_c = 3 \times 10^3 \text{ A/m}$$

$$T_c = 7.2 \text{ K}$$

$$r = 0.5 \text{ mm}, I_c = ?$$

1) Formula,

i)
$$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$\frac{H_c(T)}{H_c(0)} = 1 - \frac{T^2}{T_c^2}$$

$$\frac{T^2}{T_c^2} = 1 - \frac{H_c(T)}{H_c(0)}$$

$$\therefore \quad T^2 = T_c^2\left[1 - \frac{H_c(T)}{H_c(0)}\right]$$

$$\therefore \quad T^2 = (7.2)^2\left[1 - \frac{3 \times 10^3}{6.5 \times 10^3}\right]$$

$$= 51.84\,[0.54]$$

$$\therefore \quad T^2 = 27.99$$

$$\therefore \quad T = 5.29 \text{ K}$$

ii) Critical current, $I_c = 2\pi r H_c$

$$= 2 \times 3.14 \times 0.5 \times 10^{-3} \times 3 \times 10^{-3}$$

$$\therefore \quad I_c = 9.42 \ \mu\text{A}$$

iii) Critical current density,

$$J_c = \frac{I_c}{\pi r^2} = \frac{9.42}{3.14 \times (0.5 \times 10^{-3})^2}$$

$$J_c = \mathbf{12 \times 10^6 \ A/m^2}$$

Example 6.2.4 *For specimens of V_3 Ga, the critical fields are 1.4×10^5 A/m and 4.2×10^5 A/m at 14 K and 13 K respectively. Calculate the value of transition temperature.*

GTU : Summer-22, Marks 4

Solution : Given :

$$H_c(T_1) = 1.4 \times 10^5, T_1 = 14 \text{ K}$$

$$H_c(T_2) = 4.2 \times 10^5, T_2 = 13 \text{ K}$$

Formula, $$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$\therefore \quad 1.4 \times 10^5 = H_c(0)\left[1 - \left(\frac{14}{T_c}\right)^2\right] \qquad \dots (1)$$

$$\therefore \quad 4.2 \times 10^5 = H_c(0)\left[1 - \left(\frac{13}{T_c}\right)^2\right] \qquad \dots (2)$$

(1) ÷ (2)

$$\frac{1.4 \times 10^5}{4.2 \times 10^5} = \frac{1 - \left(\frac{14}{T_c}\right)^2}{1 - \left(\frac{13}{T_c}\right)^2}$$

$$0.3333\left[1 - \left(\frac{13}{T_c}\right)^2\right] = 1 - \left(\frac{14}{T_c}\right)^2$$

$$\therefore \quad 0.3333 - 0.3333\left(\frac{13}{T_c}\right)^2 = 1 - \left(\frac{14}{T_c}\right)^2$$

$$\therefore \quad \left(\frac{14}{T_c}\right)^2 - 0.3333\left(\frac{13}{T_c}\right)^2 = 1 - 0.3333$$

$$\frac{139.6723}{T_c^2} = 0.6667$$

$$\therefore \quad T_c^2 = \frac{139.6723}{0.6667}$$

$$T_c^2 = 209.49$$

$$\therefore \quad T_c = \mathbf{14.474\ K}$$

Example 6.2.5 *The critical temperature of Hg with isotopic mass 199.5 at 4.185 K. Calculate its critical temperature when its isotopic mass changes to 203.4.*

GTU : Summer-22, 23, Marks 3

Solution :

Given : $M_1 = 199.5$ $T_{c1} = 4.185\ K$

$M_2 = 203.4$ $T_{c2} = ?$

Formula, $T_c \cdot M^{\frac{1}{2}} = \text{Constant}$

$$\frac{T_{c1}}{T_{c2}} = \frac{\sqrt{M_1}}{\sqrt{M_2}}$$

$$\therefore \quad T_{c2} = \frac{T_{c1} \times \sqrt{M_1}}{\sqrt{M_2}} = \frac{4.185 \times \sqrt{199.5}}{\sqrt{203.4}}$$

$$= \frac{59.1108}{14.26183}$$

$$\therefore \quad T_{c2} = \mathbf{4.144\ K}$$

Example 6.2.6 *Calculate the critical current for a wire of Pb having a diameter of 1 mm at 4.5 K. The critical temperature of Pb is 7.2 K and $H_c(0) = H(0) = 6.5 \times 10^4$ A/m.*

GTU : Summer-23, Marks 3

Solution : Given : $d = 1$ mm, $T = 4.5$ K

$T_c = 7.2$ K, $H_c(0) = H(0) = 6.5 \times 10^4$ A/m

Formula,
$$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$H_c = 6.5 \times 10^4\left[1 - \left(\frac{4.5}{7.2}\right)^2\right] = 6.5 \times 10^4 \times 0.61$$

$$H_c = 3.965 \times 10^4\ \text{A/m}$$

$\therefore$ Critical current,

$$I_c = 2\pi r H_c = 2 \times 3.14 \times 0.5 \times 10^{-3} \times 3.965 \times 10^4$$

$$= 12.4501 \times 10^1$$

$$\therefore \quad I_c = \mathbf{124.5\ A}$$

Example 6.2.7 *Calculate the critical current which can flow through a long thin superconding wire of diameter 10^{-3} m.*

GTU : Summer-23, Marks 4

Solution :

Given : $H_c = 7.9 \times 10^3$ A/m, $d = 10^{-3}$ m

Formula,
$$I_c = 2\pi r H_c = 2 \times 3.14 \times 0.5 \times 10^{-3} \times 7.9 \times 10^3$$

$$= 24.806$$

$$I_c = \mathbf{24.806\ A}$$

Example 6.2.8 *The critical current density equal to 1.71×10^8 A/m^2 is required to change a superconding wire to radius 0.5 mm at 4.18 K. If the critical temperature of the material is 7.5 K, Calculate the maximum value of the critical magnetic field.*

GTU : Winter-21, Marks 3

Solution : Given : $J_c = 1.71 \times 10^8$ A/m^2, r = 0.5 mm,

$$T = 4.18 \text{ K}, T_c = 7.5 \text{ K}, H_c = ?$$

Formula, $$J_c = \frac{I_c}{\pi r^2}, I_c = 2\pi r H_c$$

$$\therefore \quad J_c = \frac{I_c}{\pi r^2}$$

$$\therefore \quad I_c = J_c \times \pi r^2 = 1.71 \times 10^8 \times 3.14 \times (0.5 \times 10^{-3})^2$$

$$I_c = 1.34235 \times 10^2 \text{ A}$$

$$\therefore \quad I_c = \mathbf{134.235\ A}$$

$$I_c = 2\pi r H_c$$

$$\therefore \quad H_c = \frac{I_c}{\pi r^2} = \frac{134.235}{2 \times 3.14 \times 0.5 \times 10^{-3}}$$

$$H_c = 42.75 \times 10^3$$

$$\therefore \quad H_c = \mathbf{4.275 \times 10^4\ A/M}$$

Example 6.2.9 *Calculate the critical current for a superconducting wire of lead having a diameter of 1 mm at 4.2 K critical temperature for lead is 7.18 K and $H_c(0) = 6.5 \times 10^4$ A/m.*

GTU : Winter-23, Marks 4, Winter-21 Marks 3

Solution : Given : d = 1 mm, T = 4.2, T_c = 7.18 K

$H_c(0) = 6.5 \times 10^4$ A/m, I_c = ?

Formula, 1) $H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$

2) $I_c = 2\pi r H_c$

1) $$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$= 6.5 \times 10^4 \left[1 - \left(\frac{4.2}{7.18}\right)^2\right]$$

$$H_c(T) = 4.276 \times 10^4 \text{ A/m}$$

2) Critical current, $$I_c = 2\pi r H_c$$

$$= 2 \times 3.14 \times 0.5 \times 10^{-3} \times 4.276 \times 10^4$$

$$\therefore \quad I_c = \mathbf{134.39\ A}$$

Example 6.2.10 *The magnetic field intensity in the tin material is $\frac{3 \times 10^5}{4\pi}$ A/m at 0 K. Calculate the temperature of the superconductor the field intensity is measured as $\frac{2 \times 10^5}{4\pi}$ A/m. Critical temperature of tin is 3.69 K.*

GTU : Winter-22, Marks 4

Solution :

Given :

$$H_c(0) = \frac{3 \times 10^5}{4\pi} \text{ A/m},$$

$$H_c(T) = \frac{2 \times 10^5}{4\pi} \text{ A/m}, \qquad T = ?$$

$$T_c = 3.69 \text{ K}$$

Formula, $$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$\therefore \quad \frac{H_c(T)}{H_c(0)} = 1 - \frac{T^2}{T_c^2}$$

$$\therefore \quad \frac{T^2}{T_c^2} = 1 - \frac{H_c(T)}{H_c(0)}$$

$$\therefore \quad T^2 = T_c^2\left[1 - \frac{H_c(T)}{H_c(0)}\right] = (3.69)^2\left[1 - \frac{2 \times 10^5}{3 \times 10^5}\right]$$

$$= 13.6161\ [0.3334]$$

$$\therefore \quad T^2 = 4.539$$

$$\therefore \quad T = \mathbf{2.13\ K}$$

Example 6.2.11 *i) The critical field for a superconducting sample is 10^6 A/m at 9.58 K and 2×10^6 A/m at 0 K determine the T_c value.*

ii) Calculate the critical current through a long thin superconducting wire to diameter 0.5 mm. The critical field is 9.2 kA/m.

GTU : Winter-22, Marks 7

Solution :

i) $$H_c(T) = 10^6 \text{ A/m}, T = 9.58 \text{ K}$$

$$H_c(0) = 2 \times 10^6, \quad T_c = ?$$

Formula, $$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$\therefore \quad \frac{H_c(T)}{H_c(0)} = 1 - \frac{T^2}{T_c^2}$$

$$\frac{T^2}{T_c^2} = 1 - \frac{H_c(T)}{H_c(0)}$$

$$\therefore \quad T_c^2 = \frac{T^2}{1 - \frac{H_c(T)}{H_c(0)}}$$

$$\therefore \quad = \frac{(9.58)^2}{1 - \frac{1 \times 10^6}{2 \times 10^6}}$$

$$= \frac{91.7764}{0.5}$$

$$T_c^2 = 183.5528$$

$$T_c = \mathbf{13.54\ K}$$

$\therefore$

ii) Given :

$$d = 0.5\ \text{mm} = 0.5 \times 10^{-3}\ \text{m}$$

$$H_c = 9.2\ \text{kA/m} = 9.2 \times 10^3\ \text{A/m}$$

$$I_c = ?$$

$$\text{Critical current, } I_c = 2\,\pi\, r\, H_c$$

$$= 2 \times 3.14 \times 0.25 \times 10^{-3} \times 9.2 \times 10^3$$

$$I_c = \mathbf{14.44\ A}$$

Example 6.2.12 *The critical magnetic field of lead wire 6.5×10^3 A/m at 0 K. At what temperature would the critical magnetic field of lead drop to 4.5×10^3 A/m, if the critical temperature of lead is 7.18 K ? What is the critical current density at that temperature if the diameter of the wire is 2×10^{-3} m ?* **GTU : Winter-23, Marks 4**

Solution :

Given :

$$H_c(0) = 6.5 \times 10^3\ \text{A/m},$$

$$H_c(T) = 4.5 \times 10^3\ \text{A/m},\ T = ?$$

i) $T_c = 7.18$ K

ii) $J_c = ?,\ d = 2 \times 10^{-3}$ m

i)

$$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]^2$$

$$\therefore \quad \frac{H_c(T)}{H_c(0)} = 1 - \frac{T^2}{T_c^2}$$

$$\therefore \quad \frac{T^2}{T_c^2} = 1 - \frac{H_c(T)}{H_c(0)}$$

$$\therefore \quad T^2 = T_c^2\left[1 - \frac{H_c(T)}{H_c(0)}\right]$$

$$= (7.18)^2\left[1 - \frac{4.5 \times 10^3}{6.5 \times 10^3}\right]$$

$$= 51.5524\,[0.3077]$$

$$T^2 = 15.8626$$

$$\therefore \quad T = \mathbf{3.9827\ K}$$

ii) Critical current.

$$I_c = 2\pi r H_c$$

$$= 2 \times 3.14 \times 1 \times 10^{-3} \times 4.5 \times 10^{-3}$$

$$I_c = \mathbf{28.26\ A}$$

iii) Critical current density.

$$J_c = \frac{I_c}{\pi r^2}$$

$$= \frac{28.26}{3.14 \times (1 \times 10^{-3})^2}$$

$$J_c = \mathbf{9 \times 10^6\ A/m^2}$$

Example 6.2.13 *Calculate the critical current through a long thin superconducting wire of radius 10^{-3} m. The critical magnetic field is 6.2×10^3 A/m.* **GTU : Winter-23, Marks 3**

Solution : Given : $r = 10^{-3}$ m, $H_c = 6.2 \times 10^3$ A m.

Formula,

$$I_c = 2\pi r H_c$$

$$= 2 \times 3.14 \times 10^{-3} \times 6.2 \times 10^3$$

$$I_c = \mathbf{38.936\ A}$$

Example 6.2.14 *The critical temperature of the superconductor is 7.5 K. Calculate the critical field at 5 K. At 0 K the critical field is 0.24 T.* **GTU : Winter-22, Marks 3**

Solution : Given : $T_c = 7.5$ K, $H_c(T) = ?$ $T = 5$ K.

$$H_c(0) = 0.24\ T$$

Formula,

$$H_c(T) = H_c(0)\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

$$= 0.24\left[1 - \left(\frac{5}{7.5}\right)^2\right]$$

$$= 0.24\,[0.5556]$$

$$H_c(T) = \mathbf{0.133\ K.}$$